



SGGS Institute of Engineering and Technology, Nanded

Summer Term Examination 2024-25

TY B Tech

Power Electronics

Instrumentation Engineering

PCC-IN310

Date: 17-Jun-25 Time: 10.00 To 13.00

Max. Marks: 100

Notes:

1. All questions are compulsory.
2. Use of non programmable calculator is permitted.
3. This paper contains one page.
4. Assume suitable data, if necessary.
5. Figures to right indicate full marks.

Marks

Q.1 Attempt any four:

- a Define (i) Holding current, (ii) Duty cycle, (iii) Firing angle, (iv) Displacement factor and (v) Overlap angle. (5)
- b Describe the different modes of operation of a TRIAC with the help of its V-I characteristics and discuss its applications. (5)
- c Explain the operation of a single phase PWM inverter. How is the RMS value of output voltage varied? (5)
- d What are the various means for speed control of 3-phase induction motor? Explain stator voltage control scheme in brief. (5)
- e What do you mean by Snubber circuit? Draw the circuit and explain the function of each component. (5)

- Q.2 a Describe the operation of single-phase, fully controlled rectifier without freewheeling diode and derive an expression for its average output voltage. (10)
- b Explain the procedure for designing various elements of a UJT relaxation oscillator used for SCR triggering. (10)

- Q.3 a Explain the operation of a single-phase AC voltage controller supplying a resistive load and controlled by the ON/OFF controlled method of control. (10)
- b With the help of circuit diagram and waveforms, explain the operation of step-down chopper and derive an expression for its output voltage. (10)

- Q.4 a With the help of circuit diagram and relevant waveforms, explain the operation of a three-phase half wave-controlled rectifier. (10)
- b Describe the operation of a three-phase inverter with 120° conduction mode and star connected three phase balanced resistive load. (10)

- Q.5 a A dc separately excited motor is powered by dc chopper from 250 V DC source. The armature resistance is 0.04Ω and the back emf constant of the motor is 1.3 V/A rad/s . The average armature current is 80 A and the field current is constant at 2 A. If the duty cycle of the chopper is 60 %, determine (i) the motor speed, (ii) the developed torque. (10)
- b With the help of neat circuit diagram and relevant waveforms, describe the operation of single phase to single phase cycloconverter. (10)

//End of Paper//



SGGS Institute of Engineering and Technology, Vishnupuri, Nanded
Academic Year 2024-25 (Semester I)

Summer Term Examination

Subject: PCC-IN 303 Digital Signal Processing
 Class: T. Y. (Instrumentation Engg.)

Date: 16/06/2025

Time: 10.00 -13.00 hrs

Max. Marks : 100

Note: All questions are compulsory

QN		Question	Marks
Q. 1	a	Obtain the z-transform of. (i) $x[n] = a^n u[n-3]$ (ii) $x[n] = \cos(\omega_0 n) u[n]$	10
	b	Input-Output relationship of a causal LTI system is $y[n] = -\frac{7}{12} y[n-1] - \frac{1}{12} y[n-2] + x[n]$; Determine the transfer function of the system with ROC. Also determine the impulse response of the system.	10
Q.2	a	Suppose $X(z)$, the z-transform of a causal sequence $x[n]$, is $X(z) = \frac{1}{(z - \frac{1}{4})(z - \frac{1}{2})}$ Determine ROC of $X(z)$ and $x[n]$ at $n=1$	10
	b	Let the transfer function of an LTI system is $H(z) = 1 + 2z^{-1} + 3z^{-2} + 2z^{-3}$ Determine the output $y[n]$ of the system for following input $x[n]$. $x[n] = \{2, 1, 3, 1, 2, 3\}$ $\uparrow n=0$ Also indicate the $y[0]$ sample of the output.	10
Q. 3	a	Compute DFT of the following sequence using Decimation-In-Frequency (DIF) FFT algorithm $x[n] = \{1, 2, 3, 2\}$ What is bit reversed order?	10
	b	Using DFT property, compute the circular convolution of the following periodic sequences with period $N = 4$. $\tilde{x}[n] = \{3, 2, 1, 2\}$ and $\tilde{h}[n] = \{1, 1, 3, 2\}$.	10
Q. 4	a	Assume the constraints on IIR Digital filter $0.8 \leq H(e^{j\omega}) \leq 1 \quad 0 \leq \omega \leq 0.2\pi$ $ H(e^{j\omega}) \leq 0.2 \quad 0.6\pi \leq \omega \leq \pi$ and $T=1s$. Design an analog filter (obtain $H(s)$) using Butterworth approximation. Assume Impulse	10

		Invariance Technique (IIT) for analog to digital filter transformation.	
	b	Design a non-recursive (FIR) low-pass filter of length $N=5$ for the following specifications $\left H(e^{j\omega}) \right = \begin{cases} 1, & 0 \leq \omega \leq 0.6\pi \\ 0, & 0.6\pi \leq \omega \leq \pi \end{cases}$ Use Bartlett Triangular Window to reduce the Gibb's phenomenon and obtain windowed causal filter impulse response.	10
Q. 5	a	Let the input-output relationship of a system is $y[n] = 2x[n] + x[n-1]$. Test the system for Linearity, Time-Invariance, Causality, Memoryless and stability	10
	b	The impulse response of a causal LTI system is $h[n] = (\frac{1}{2})^n u[n]$ Determine the transfer function of the system and the output of the system for input $x[n] = u[n]$	10

The End

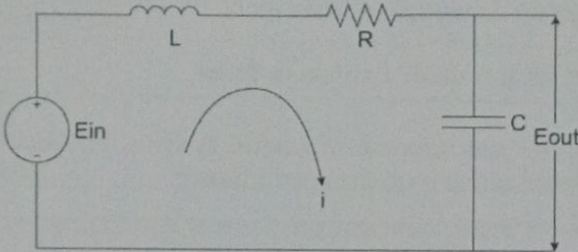
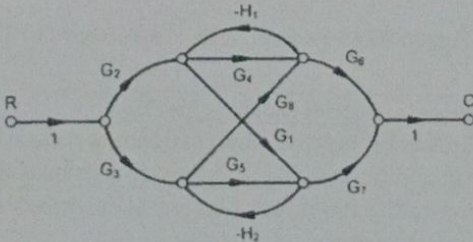
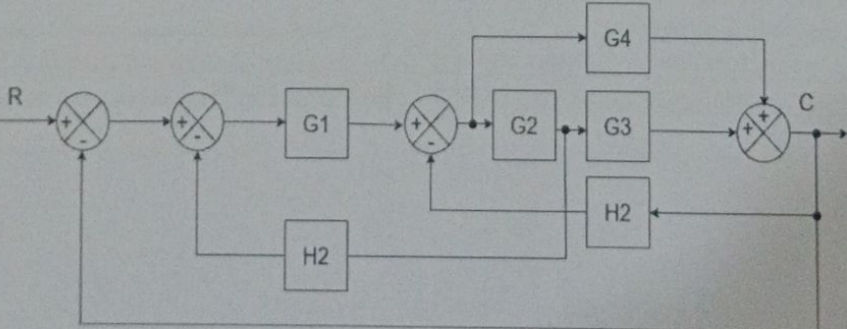


Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded.
Department of Instrumentation Engineering
Summer Term Examination
Academic Year: 2024-25

Class: TY B.Tech	Subject: Feedback Control Systems	Subject Code: PCC-IN301
Date: 18.06.2025	Time: 10:00 to 13:00 Hrs	Max. Marks: 100

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange / sharing of stationery, calculator, etc. is not allowed.
- Figures to the right indicate full marks.
- Question one (Q.1) is compulsory. Solve any eight from remaining Questions.**

Q.N.		Marks	CO	BT
Q. 1	1. Differentiate between closed loop and open loop system. 2. Write a short note on regenerative feedback. 3. Determine the transfer function of following electrical network,  4. Obtain the overall transfer function C/R from the signal flow graph shown below, 	20	CO2	BT1
Q.2	Using block diagram reduction technique, find the closed loop transfer function of system whose block diagram is given below, 	10	CO2	BT5

Q. 3	Explain step response of first order and second order system.	10	CO2	BT2
Q.4	What do you mean by stability? Explain response of system on the basis of various pole placement in s plane.	10	CO4	BT4
Q. 5	Find out the following systems are stable or not using Routh array method, 1. $s^4 + 8s^3 + 18s^2 + 16s + 5 = 0$ 2. $3s^4 + 10s^3 + 5s^2 + 5s + 2 = 0$	10	CO3	BT2
Q.6	Explain the root locus concept in details.	10	CO3	BT2
Q. 7	Consider the unity feedback control system with open loop transfer function, $G(s) = \frac{K(s+1)(s+2)}{(s+0.1)(s-1)}$ Draw the root loci of the system with gain K as variable. As an aid to plotting determine; asymptotes, centroid, breakaway point, the gain at which root locus crosses j ω axis.	10	CO3	BT6
Q.8	Write a short note on lead lag compensator.	10	CO4	BT5
Q.9	Explain general procedure for constructing Bode plots.	10	CO3	BT2
Q.10	What is Polar plot? What for it is used? Explain in detail.	10	CO6	BT3
Q.11	Discuss basic control actions and response of control systems, effects of integral and derivative control action on system performance.	10	CO1	BT2

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	To remember the innovative ideas to improve plant efficiency reduce leakages and losses and use technologies for designing and developing pollutant free industrial environment.
CO 2	To understand the operation of traditional power plants and describe the instruments that makes up their measurement and control systems.
CO 3	To apply the knowledge of mathematics for deriving mathematical models for different processes in the power plant.
CO 4	To analyse the various instruments used in power plant and make recommendations for improving the control processes.
CO 5	To evaluate the performance of boilers and turbines.
CO 6	To design instrumentation systems for electricity generating plants.